

WhatsApp NLP & Conversation Analytics Dashboard

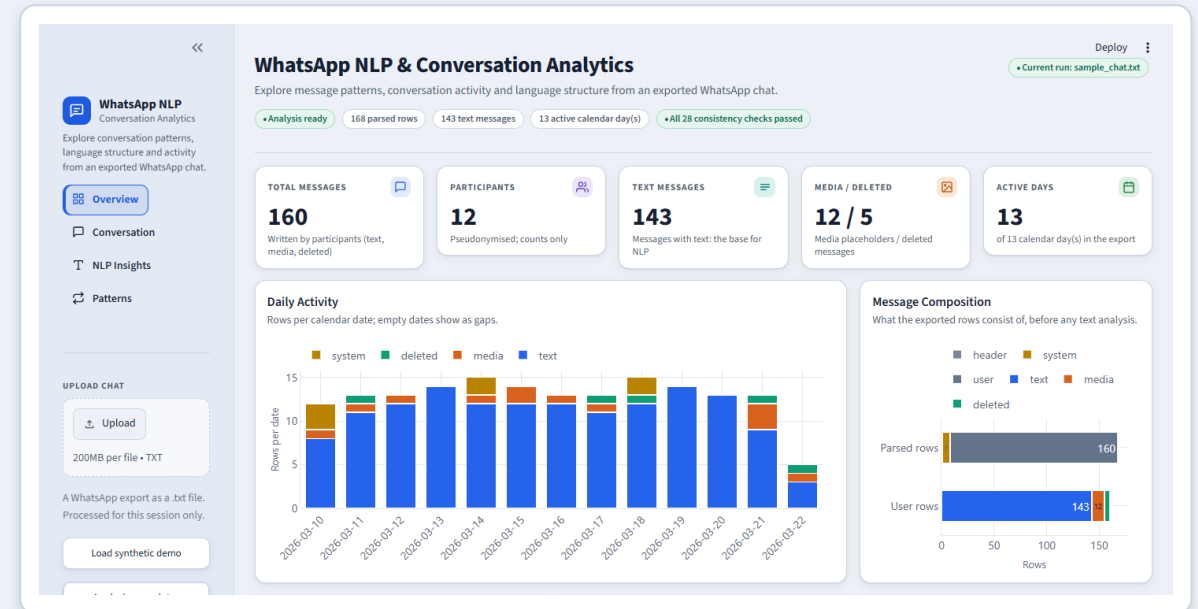
A privacy-aware pipeline and interactive dashboard for describing the language and activity structure of exported WhatsApp chats

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Overview page, fictional sample chat

Problem & Objectives

Why a raw chat export needs a careful, explainable analysis pipeline

PROBLEM

An exported WhatsApp chat is a raw text file. It is hard to see directly:

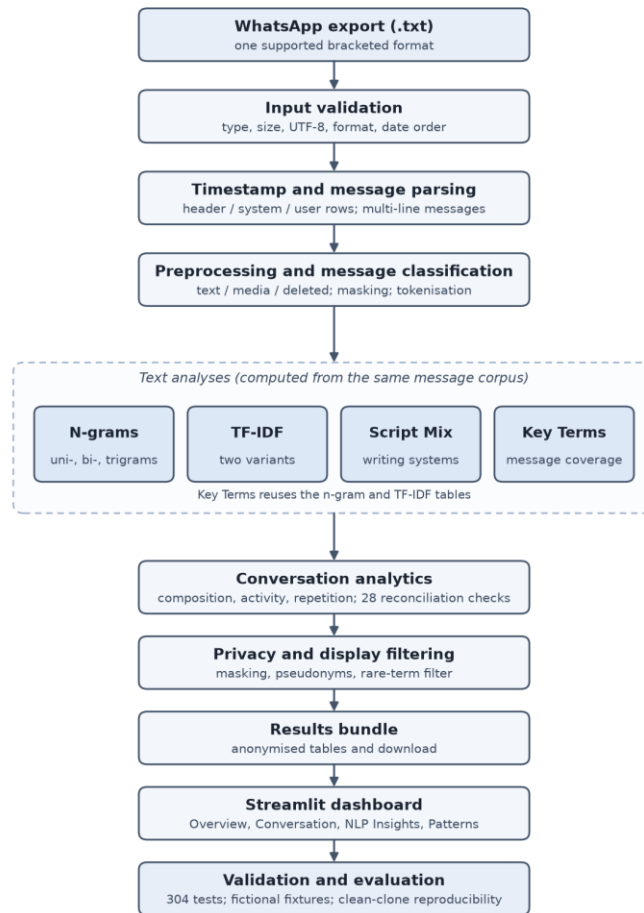
- Message patterns
- Frequently used terms
- Activity over time
- Repeated content
- Writing-system usage
- Conversation composition

OBJECTIVES

- 1 Build a robust WhatsApp parser and preprocessing pipeline
- 2 Apply NLP techniques: n-grams and TF-IDF
- 3 Analyse Script Mix and key conversational terms
- 4 Extract activity and repetition patterns
- 5 Present results in a privacy-conscious interactive dashboard

Data & Methodology

One supported export format, one reproducible pipeline



WHAT HAPPENS TO AN UPLOADED CHAT

1

Validate and parse

Bracketed .txt export; header, system and user rows; multi-line messages kept together

2

Preprocess and classify

Text, media or deleted; links and phone numbers masked; Unicode-aware tokens

3

Analyse

N-grams, TF-IDF, Script Mix, Key Terms and conversation analytics (28 reconciliation checks)

4

Filter and present

Privacy filter, anonymised results bundle, Streamlit dashboard

Public demonstration uses a fictional WhatsApp-style dataset for privacy and reproducibility.

NLP Analysis

N-GRAMS

Unigrams

Bigrams

Trigrams

All messages vs unique texts

TF-IDF

Term distinctiveness

Primary + unique-text variants

Validated against scikit-learn

SCRIPT MIX

Latin

Devanagari

Mixed-script

No-alphabetic

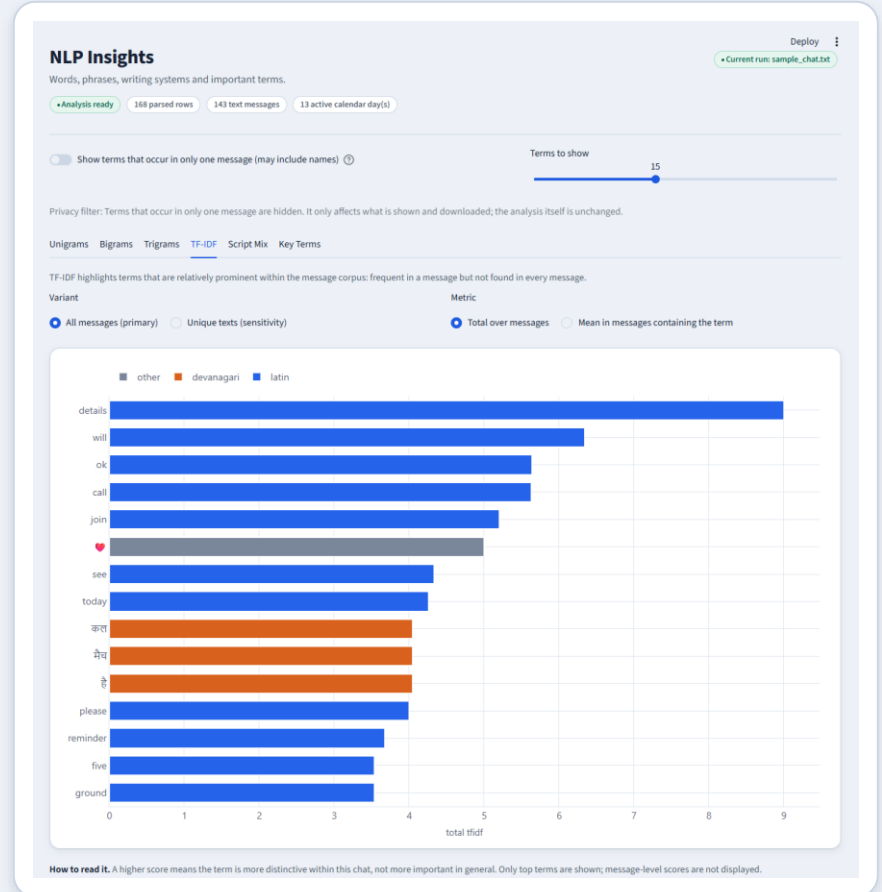
KEY TERMS

Message coverage

Total occurrences

Repeated-message effect

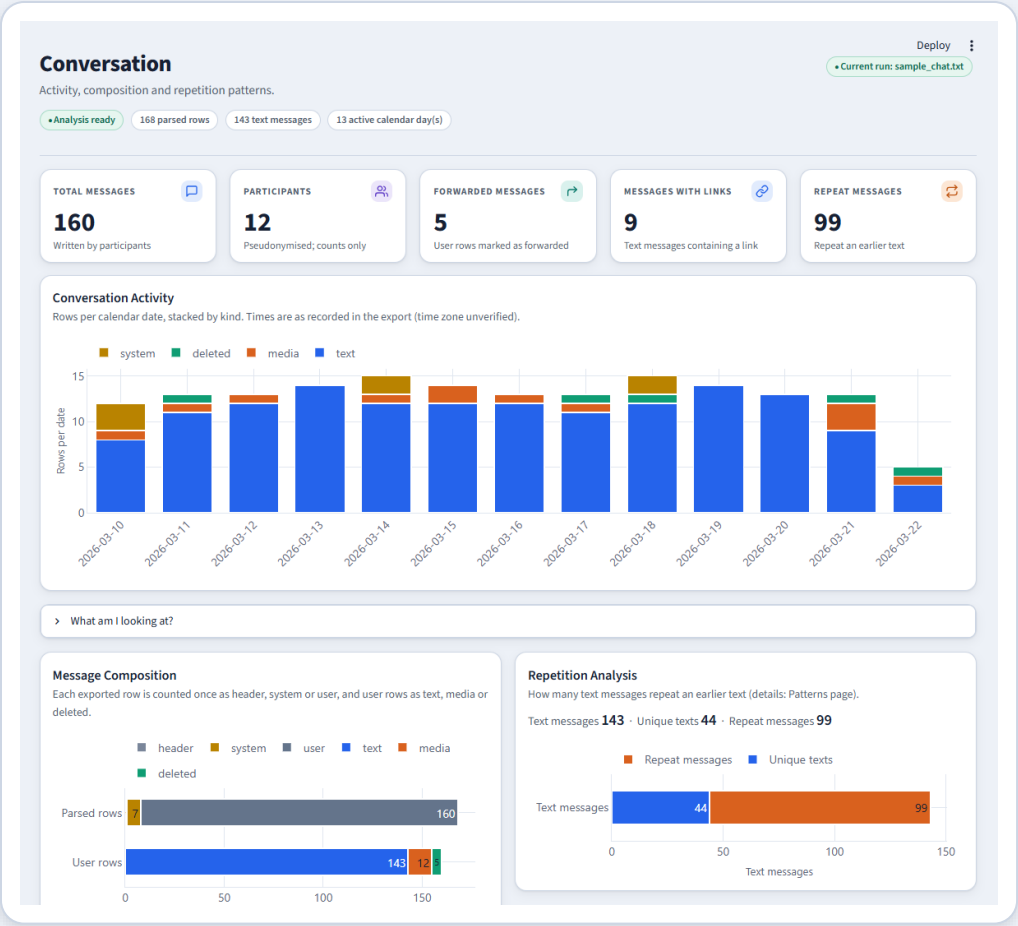
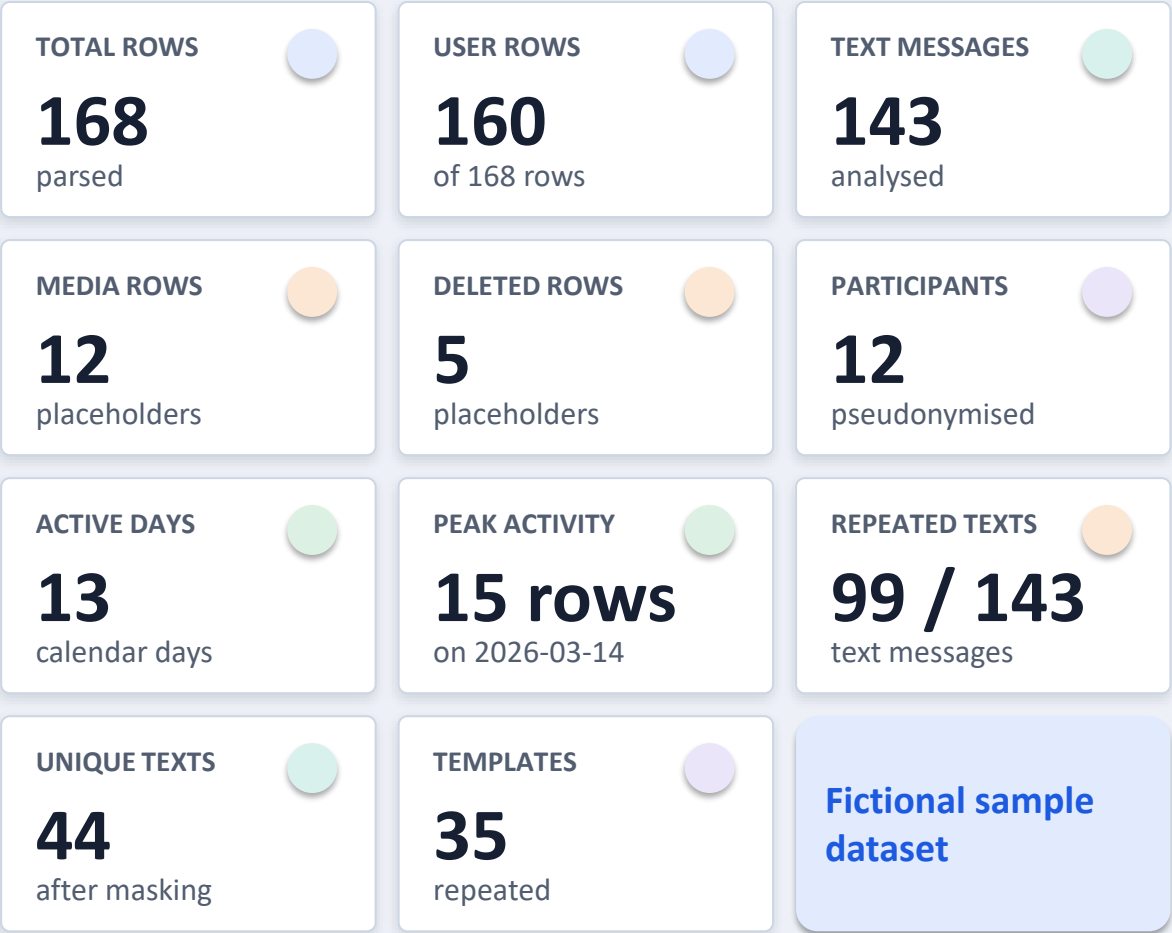
Script Mix is a writing-system analysis, not language identification.



TF-IDF tab, fictional sample chat

Conversation Analytics

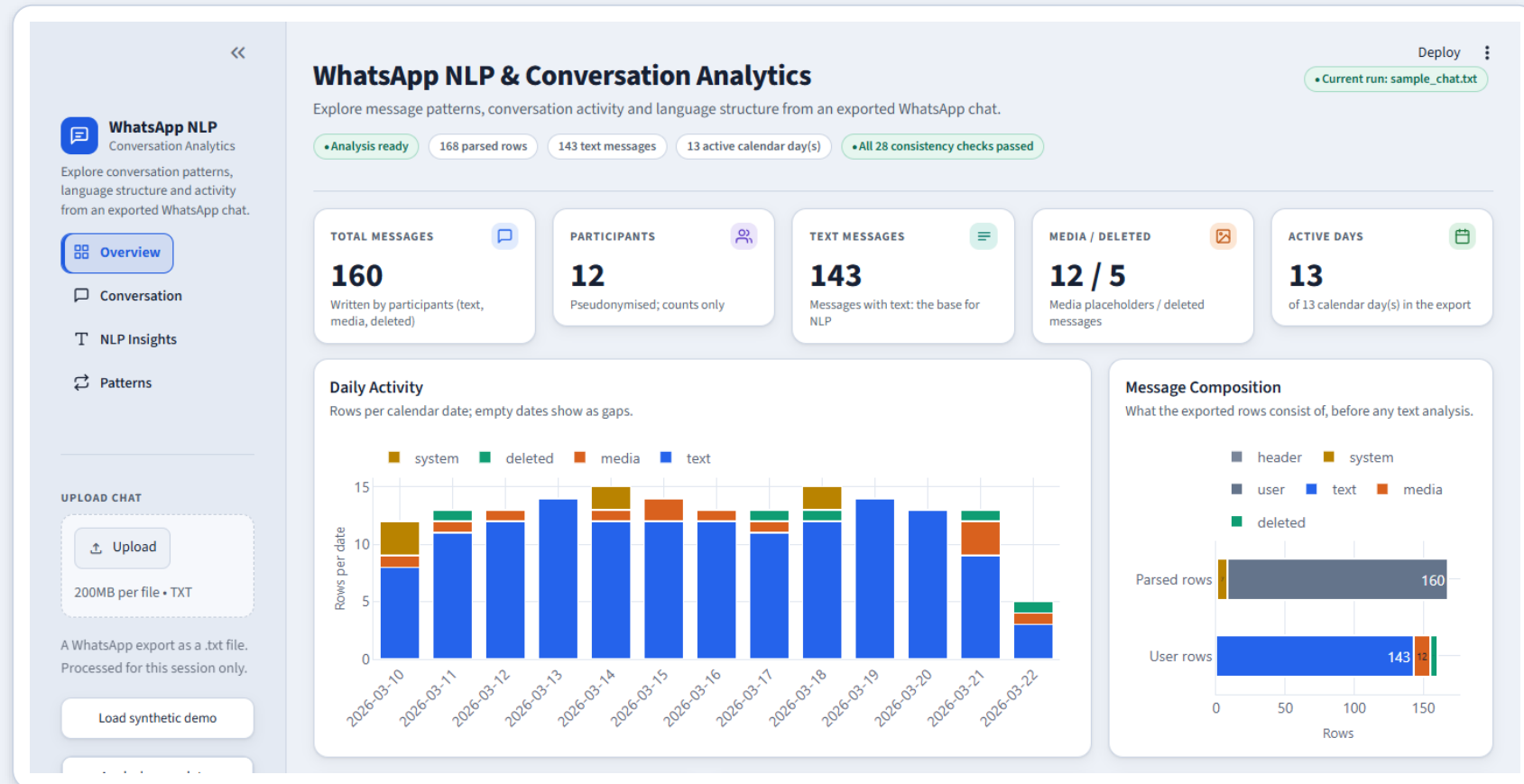
Fictional sample dataset: 12 invented participants



Conversation page, fictional sample chat

Interactive Dashboard

Streamlit + Plotly academic prototype; upload-driven, session-based analysis



Overview

key figures, daily activity, composition, key findings

Conversation

activity, composition, repetition, hour and weekday tables

NLP Insights

six tabs: n-grams, TF-IDF, Script Mix, Key Terms

Patterns

repetition summary and anonymous templates

Overview page loaded with the fictional sample chat (12 invented participants)

Privacy-conscious display

Anonymised download

Optional theme image

Validation & Limitations

What was checked, and what the results do not show

VALIDATION

- 304 tests passed in a clean clone
- 308 tests locally (4 are local-only)
- 28 reconciliation checks
- TF-IDF vs scikit-learn: max difference about 1.1×10^{-16}
- Committed outputs reproduced

PRIVACY

- Fictional public dataset
- Private chat excluded from the repository
- Privacy and display filtering
- Repository privacy scan
- No private chat results shown publicly

LIMITATIONS

- Supported export format is limited
- Android / no-bracket formats unsupported
- Script Mix is not language identification
- Privacy filter is not complete PII detection
- Public results are not population-level findings

Conclusion & Future Work

CONCLUSION

A reproducible pipeline for analysing WhatsApp-style exported conversations, with its limits stated.

- Preprocessing, n-grams and TF-IDF
- Script Mix and Key Terms
- Conversation analytics and an interactive dashboard
- Validation shows reproducibility and robustness
- A fictional public dataset enables safe demonstration

FUTURE WORK

- 1 Broader WhatsApp export-format support
- 2 Larger, real-world evaluation (group and iOS exports)
- 3 Richer multilingual analysis and stopwords handling
- 4 Stronger personal-data detection
- 5 More robust large-scale TF-IDF evaluation

Thank you

Questions?